

Homework 8

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Problem 1a.

Proof. $\forall \alpha \in \mathbb{R}$:

1. Case 1: $\alpha \leq 0$. Let $K = 1, \forall n \geq K, \sqrt{n} > 0 \geq \alpha$.
2. Case 2: $\alpha > 0$. By the Archimedean property, $\exists K \geq \alpha^2$. Thus $\sqrt{K} > \sqrt{\alpha^2 + 1} > \sqrt{\alpha^2} \geq \alpha$. Since $\forall n \geq K, \sqrt{n} \geq \sqrt{K}$ and since $\sqrt{K} > \alpha$, thus $\sqrt{n} > \alpha$.

Thus $\sqrt{n}$ is properly divergent and $\lim_{n \rightarrow \infty} \sqrt{n} = +\infty$ □

Problem 1b.

Proof. $\forall n \in \mathbb{N} \ni n > 1, n^2 > n \implies n^2 - n > 0$.

$\forall \alpha \in \mathbb{R}$:

1. Case 1: $\forall \alpha \leq 0$. Let $K = 2, \forall n \geq K, n \geq K > 1$ so $n^2 - n > 0 \geq \alpha \implies n^2 - n \geq \alpha$.
2. Case 2: $\alpha > 0$. By the Archimedean property, $\exists K > \frac{1+\sqrt{1+4\alpha}}{2}$. $K^2 - K > \left(\frac{1+\sqrt{1+4\alpha}}{2}\right)^2 - \left(\frac{1+\sqrt{1+4\alpha}}{2}\right) = \frac{1}{4}(1 + 2\sqrt{1+4\alpha} + 1 + 4\alpha) - \left(\frac{1+\sqrt{1+4\alpha}}{2}\right) = \alpha$. Thus $K^2 - K > \alpha$. $\forall n \geq K, n^2 \geq nK \geq K^2 \implies n^2 - n \geq K^2 - K$. Since $K^2 - K > \alpha$ and $n^2 - n \geq K^2 - K, n^2 - n > \alpha$.

□

Problem 1c.

Proof. $\forall c \in \mathbb{R}$.

If $c \leq 0$, then let $n = 2$. Thus $(-1)^n n = 2 > 0 \leq c \implies (-1)^n n > c$. Thus c is not an upper bound for $(-1)^n n$.

Otherwise if $c > 0, \forall K \geq \lceil c \rceil + 1$, thus $K > c$. If $K \bmod 2 = 1$, then $L = K + 1$ is even. Thus $(-1)^L L = L > K > c$. If $K \bmod 2 = 0$, thus $(-1)^K K = L > c$. Thus c is not an upper bound for $(-1)^n n$.

Thus $((-1)^n n)$ is unbounded and thus divergent.

Let $\alpha = 0, \forall K \in \mathbb{N}$, let $L = K + 1$:

1. Case 1 ($K \bmod 2 = 0$): $(-1)^K K = K > \alpha = 0$ and $(-1)^L L = -L < 0 = \alpha$.
2. Case 2 ($K \bmod 2 = 1$): $(-1)^L L = L > \alpha = 0$ and $(-1)^K K = -K < 0 = \alpha$

Thus $(-1)^n n$ is not properly divergent. □

Problem 2.

Proof. $\Rightarrow$: Suppose (x_n) is a sequence where $\forall n \in \mathbb{N} \ni x_n > 0$ and $\lim_{n \rightarrow \infty} x_n = 0$. Since $x_n > 0$, then $\frac{1}{x_n} > 0$. $\forall \alpha \in \mathbb{R} \ni \alpha \leq 0$, $K = 1$ and $\forall n \geq K \alpha \leq 0 < \frac{1}{x_n}$.

$\forall \alpha \in \mathbb{R} \ni \alpha > 0$. Let $\epsilon = \frac{1}{\alpha}$. By definition of limit, $\exists K \in \mathbb{N} \ni \forall n \geq K, \exists |x_n| < \epsilon$. Since $x_n > 0$, $x_n < \epsilon \Rightarrow \frac{1}{\epsilon} < \frac{1}{x_n}$. Thus $\alpha < \frac{1}{x_n}$.

Thus $(\frac{1}{x_n})$ is properly divergent and $\lim_{n \rightarrow \infty} (x_n) = \infty$

$\Leftarrow$: Suppose (x_n) is a sequence where $\forall n \in \mathbb{N} \ni x_n > 0$ and $\lim_{n \rightarrow \infty} \frac{1}{x_n} = \infty$.

$\forall \epsilon > 0$. Let $\alpha = \frac{1}{\epsilon}$. By definition of proper divergence, $\exists K \in \mathbb{N} \ni \forall n \geq K, \frac{1}{x_n} > \alpha$. Since $x_n > 0$, $|x_n| = x_n$. So $\frac{1}{|x_n|} > \alpha$. Thus $\epsilon = \frac{1}{\alpha} > |x_n|$. Thus $|x_n| < \epsilon$.

Thus by definition of convergence, $\lim_{n \rightarrow \infty} x_n = 0$. □

Problem 3.

Proof. Let $S_n = \sum_{k=0}^n \frac{1}{n(n+1)}$.

$$\begin{aligned} S_n &= \sum_{k=1}^n \frac{1}{n(n+1)} = \sum_{k=1}^n \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= \sum_{k=1}^n \frac{1}{n} - \sum_{k=1}^n \frac{1}{n+1} \\ &= \sum_{k=1}^n \frac{1}{n} - \sum_{k=2}^{n+1} \frac{1}{n} \\ &= 1 + \frac{1}{n+1} \end{aligned}$$

Since 1 is a constant sequence, it converges to 1.

$\forall \epsilon > 0$. By the Archimedean Property, $\exists K \in \mathbb{N} \ni K > \frac{1}{\epsilon} \Rightarrow \epsilon > \frac{1}{K}$.

Thus $\forall n \geq K$,

$$\begin{aligned} \left| \frac{1}{n+1} \right| &= \frac{1}{n+1} \\ &< \frac{1}{n} \quad (\text{since } n < n+1) \\ &\leq \frac{1}{K} \quad (\text{since } n \geq K) \\ &< \epsilon. \end{aligned}$$

Thus $\lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$

Thus by operation of limits, $\lim_{n \rightarrow \infty} S_n = 1$.

$\sum_{k=0}^{\infty} \frac{1}{n(n+1)}$ converges to 1. □

Problem 4.

Proof. $\implies$: Suppose (x_n) is convergent. Let $L = \lim x_n$. Since 1 is constant so convergent to 1, and $\lim \left(\frac{x_n}{y_n} \right) = r > 0$, by division of limits, $\lim \frac{y_n}{x_n} = \lim \left(\frac{1}{\frac{x_n}{y_n}} \right) = \frac{1}{\lim \frac{x_n}{y_n}} = \frac{1}{r}$. $y_n = x_n \frac{y_n}{x_n}$. By multiplication of limits, since x_n is convergent and $\frac{y_n}{x_n}$ is convergent, y_n is convergent and $\lim y_n = \lim x_n \frac{y_n}{x_n} = (\lim x_n) \left(\lim \frac{y_n}{x_n} \right) = L * \frac{1}{r} = \frac{L}{r}$.

($\Leftarrow$): Suppose y_n is convergent. Thus $\lim_{n \rightarrow \infty} y_n = L$ for $L \in \mathbb{R}$. $x_n = y_n * \frac{x_n}{y_n}$, by multiplication of limits, since y_n is convergent and $\frac{x_n}{y_n}$ is convergent, x_n is convergent and $\lim x_n = \lim \left(y_n \frac{x_n}{y_n} \right) = (\lim y_n) \left(\lim \frac{x_n}{y_n} \right) = Lr$. $\square$